# Command Guide

How to run Matrix Eye safely from VS Code · October 2026

**Short answer.** No. These commands do not touch your website's code, files or server. **npm install** only adds packages inside the matrix-eye folder. **npm run matrix -- crawl** only reads a website over the web, like a visitor would, and saves what it read to one file inside matrix-eye. The one way to cause a problem is running them from the wrong folder, so section 1 comes first.

## 1. Before you run anything: be in the right folder

Everything in this guide is run from the **matrix-eye** folder, never from your website's folder. In VS Code, use **File, Open Folder** and choose matrix-eye, or open a terminal and change into it. Then check where you are:

```
cd C:¥Users¥godca¥Downloads¥projects-longterm¥matrix-eye
dir package.json
```

**Renamed project.** The project used to be called Eagle's Eye. If your folder is still named eagle-eye, close VS Code, rename the folder to matrix-eye, and open it again; every path in this guide then matches.

You should see a package.json whose name is matrix-eye. If the terminal prompt shows any other project folder (for example your AIIT website), stop and change folder first. **Requirement:** Node.js 24 or newer. Check with node -v.

## 2. Commands at a glance

| Command | What it does | What it touches |
|---|---|---|
| npm install | Downloads the libraries Matrix Eye needs. | Creates matrix-eye/node_modules (about 100 MB). Nothing outside the folder. |
| npm test | Runs the 381 automated tests. | Starts a tiny local test server on your computer for a few seconds. No internet, no files kept. |
| npm run typecheck | Checks the code for type errors. | Reads files only. |
| npm run eval | Scores answer quality on the built-in sample university. | Local test server; writes a report to evals/reports. |
| npm run matrix -- crawl <url> | Reads the pages of a website and indexes them. | Sends read-only requests to that website; writes matrix-eye/data/matrix-eye-cli.sqlite. |
| npm run matrix -- crawl <url> --render | Same, but runs the page's JavaScript first (React/Vue sites that arrive empty). | Starts a hidden Chromium browser on your computer. Needs a one-time download. |
| npm run demo -- <url> | Reads a site, then serves a page with the real chat widget answering from it. | Sends read-only requests to that site; writes matrix-eye/data/demo.sqlite; serves http://localhost:8788 on your computer only. |
| npm run build | Builds the SDK, the chat widget and the React component. | Writes dist folders inside the packages. |
| npm run matrix -- ask "..." | Answers one question from the indexed site. | Reads the local database. No internet. |
| npm run matrix -- chat | Interactive chat that remembers follow-ups. | Reads the local database. No internet (unless you set a language model). |
| npm run matrix -- search "..." | Shows the raw passages found for a query. | Reads the local database. |
| npm run matrix -- pages | Lists every page that was indexed. | Reads the local database. |

## 3. Step by step

### Step 1. Install (once)

```
npm install
```

You may see a line about SQLite being experimental. That is a harmless Node notice. Re-running npm install later is safe.

## Step 2. Confirm everything works (optional but recommended)

```
npm test
npm run eval
```

Expect 381 passed and All Phase 1 targets met. If either fails on your machine, send me the output.

## Step 3. Index a website

Start small so you can see results quickly and keep the load on the target site tiny:

```
npm run matrix -- crawl https://your-education-site.edu --max-pages 30
```

It prints each page as it is indexed, then a summary. When it looks right, raise the limit (for example --max-pages 200) and run it again. The second run is incremental and only re-reads pages that changed.

## Step 3b. If the crawl indexes 0 pages (React / Vue / single-page sites)

Some sites send an empty page and draw everything with JavaScript, so there is nothing to read. Install the browser once, then add --render:

```
npx playwright install chromium
npm run matrix -- crawl https://your-react-site.org --max-pages 30 --render
```

Expect about 3 to 6 seconds per page, so keep --max-pages small at first. The browser blocks any attempt by the page to reach internal or private addresses.

## Step 4. Ask questions

```
npm run matrix -- ask "How much is tuition for computer science?"
npm run matrix -- ask "What is the application deadline?" --debug
npm run matrix -- chat
```

--debug shows how the answer was found. In chat, type exit to leave. Answers are quoted from the site with a source link, or the assistant says it could not find the answer. **Write down every wrong answer and every wrongly declined question.** That list is the most valuable input for improving the engine.

## Step 5. Look under the hood

```
npm run matrix -- search "scholarship deadline"
npm run matrix -- pages
```

## Step 6. Score it on your own questions (optional)

Copy evals/datasets/education-demo.json to a new file, replace the questions and expected pages with ones for your site, then run:

```
npm run eval -- --url https://your-education-site.edu --dataset my-questions.json
```

## Step 7. Get conversational answers with Ollama (nothing to configure)

Without a language model, Matrix Eye answers by quoting sentences from the site. With one, it writes a natural answer from the same passages, still with source links. **Ollama** is a free program that runs such a model on your own computer. You do **not** need an account or a login.

**1. Install Ollama once** (the Windows installer from ollama.com, or run winget install -e --id Ollama.Ollama --source winget), then **close and reopen VS Code** so the ollama command is recognised.

**2. Download the model once.** It is about 2.5 GB, Apache-2.0 licensed (fine for commercial use):

```
ollama pull qwen3:4b-instruct
```

**3. That is all.** The demo and the command-line tool look for Ollama on your computer and use it automatically, and they tell you when they do. For example, the demo prints Answers are written by qwen3:4b-instruct (found Ollama on this computer). You do not set any variables.

To check it, run npm run matrix -- llm-check. It should print OK and say the model answers directly. The first answer after a pause takes about 10 seconds while the model loads; later answers take roughly 3 to 10 seconds on a laptop with a small GPU.

If you ever want the plain quoted answers instead, add **--no-model** to a command. Only if you use a different model server (not Ollama) do you need the MATRIX_EYE_LLM_BASE_URL and MATRIX_EYE_LLM_MODEL settings; the README explains those. **Use a non-thinking model.** Models that show their reasoning (the plain qwen3:4b, DeepSeek-R1) are slower and can leak that reasoning into answers, so they are ignored automatically.

## Step 8. See the chat widget on a website

This reads a site and opens a page with the real chat window answering from it, exactly as a customer's visitors would see it. Add --render for React or Vue sites. If Ollama is installed (step 7), answers are written by the model and stream in as they are produced:

```
npm run demo -- https://www.reed.edu --max-pages 40
npm run demo -- https://aiit.network --render --max-pages 20
```

When it prints Open http://127.0.0.1:8788, open that address in your browser. Press Ctrl+C in the terminal to stop it. Everything stays on your computer. Try adding ?theme=dark, ?position=left or ?color=%231b6b4a to the address to see the styling options. The first answer after a pause can take about 10 seconds while the model loads. Add --no-model to see the plain quoted answers instead.

## Step 9. Start over (optional)

To discard everything the CLI has indexed, delete the file matrix-eye/data/matrix-eye-cli.sqlite. The demo keeps its own file, matrix-eye/data/demo.sqlite, which is replaced on every run. Nothing else is affected.

### Run the API server (later, when you build the widget)

```
copy .env.example .env
npm run dev:api
```

Edit .env first and set MATRIX_EYE_API_KEY to a long random value (16 or more characters). The server listens only on your own computer (127.0.0.1, port 8787) unless you change that.

## 4. Will these commands break my website?

| Worry | Answer |
|---|---|
| Does it change my website's code or files? | No. Matrix Eye never opens, edits or deletes your website's project files. It lives in its own folder and only reads pages over the web. |
| Does it change the live website? | No. The crawler can only send **GET** requests, the same kind a browser sends when someone visits a page. It cannot submit forms, log in, edit or delete anything. This is confirmed in the code (packages/crawler/src/fetcher.ts). |
| Does npm install affect my other projects? | No, not when run inside matrix-eye. It writes only to matrix-eye/node_modules. Run in a different project's folder by mistake, it would install into that project, which is why section 1 comes first. |
| Does crawling put load on the website? | Very little. It makes at most about 4 requests per second, two at a time, obeys the site's robots.txt and Crawl-delay, and stops at the page limit. For a small site or a shared host, start with --max-pages 30 and avoid peak hours. |
| Will it show up in the site's logs or analytics? | Yes, as visits from **MatrixEyeBot**. A firewall or security plugin (for example Cloudflare, Wordfence) may block or rate-limit it. That is expected and not harmful. If it happens, allow the bot for your own site. |
| Can it crawl my local dev server? | Not by default. For safety it refuses localhost and private addresses. Add --allow-private only for your own local site, and never for sites you do not control. |
| Where does the data go? | Only into matrix-eye/data/matrix-eye-cli.sqlite on your computer. Nothing is uploaded anywhere. If you configure a language model, the passages used for an answer are sent to that model, so choose a local one if privacy matters. |
| Is anything committed to GitHub automatically? | No. The data folder, node_modules and .env are excluded from git. Nothing is pushed unless you or I run a push. |

**Be careful with:** crawling a site you do not own or have permission to index (check its terms), running commands from the wrong folder, and sharing your .env file.

# 5. Troubleshooting

| Message or symptom | What to do |
|---|---|
| 'node' is not recognised, or version below 24 | Install Node.js 24 or newer from nodejs.org, reopen VS Code, run node -v. |
| Cannot find package.json / missing script 'matrix' | You are in the wrong folder. Go back to section 1. |
| crawl: Refusing to crawl internal host / private address | Safety guard. Use a public URL, or add --allow-private for your own local site. |
| crawl: robots.txt disallows crawling | The site asks bots to stay out. Respect that, or change the site's robots.txt if it is yours. |
| crawl finishes with 0 pages indexed | The site probably draws its content with JavaScript. Run npx playwright install chromium once, then add **--render**. If it still finds nothing, a firewall may be blocking the bot; send me the URL and output. |
| Could not start Chromium | Run npx playwright install chromium once and try again. |
| Answers say 'I couldn't find that on this website' | Either the site really does not say it, or the wording differs. Try search to see what was indexed, and send me the question. |
| Warning: SQLite is an experimental feature | Harmless. It is a Node notice, not an error. |